Constructor University

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Lab Experiment 4: Two-Port Networks

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1. Introduction

A two-port network is an electrical network with two pairs of terminals, one pair usually considered the input port and the other the output port. Two-port networks are useful because complicated circuits can be represented using matrix parameters, which describe the relationship between port voltages and port currents (**V=ZI where V=**$\begin{pmatrix} \boldsymbol{V1} \\ \boldsymbol{V2} \end{pmatrix}$ $\boldsymbol{and\ I} = \begin{pmatrix} \boldsymbol{I1} \\ \boldsymbol{I2} \end{pmatrix}$ $\boldsymbol{and\ Z} = \begin{pmatrix} \boldsymbol{Z11} & \boldsymbol{Z12} \\ \boldsymbol{Z21} & \boldsymbol{Z22} \end{pmatrix}$). In this experiment, two-port networks were analyzed using impedance parameters, admittance parameters, and transmission parameters.

For an impedance two-port network, the relationship between voltages and currents is written as:

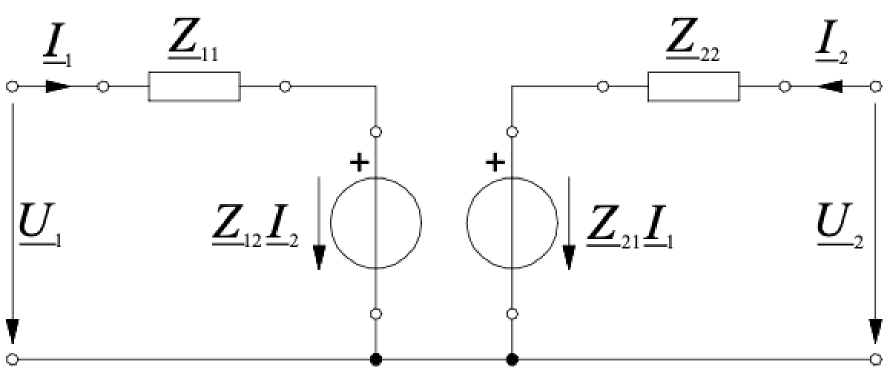
$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \cdot \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

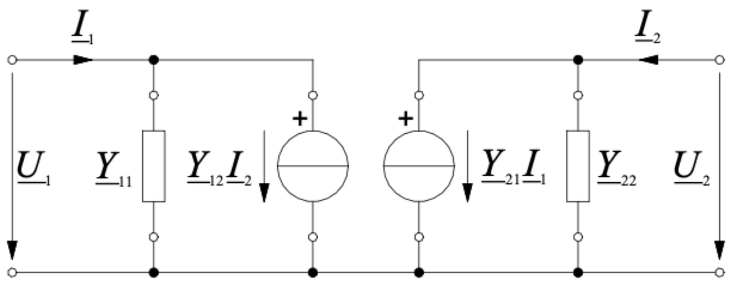
The impedance parameters can then be found by procedurally opening either I1 or I2 and measuring or calculating the current and voltage across the ends of the two-port network, then\ apply the following to find each impedance:

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}, \quad Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}, \quad Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}, \quad Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0}$$



Impedence two-port network model of a circuit.

For an admittance two-port network, the relationship is written as:

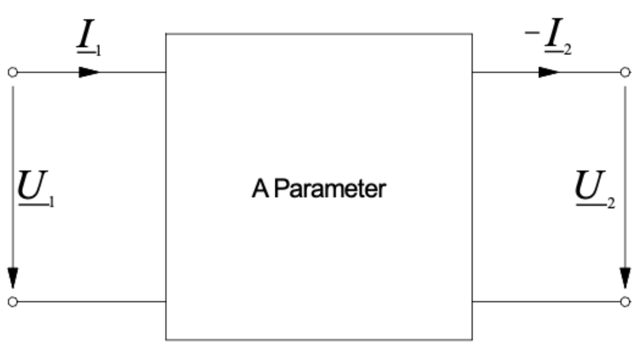


$$Y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2=0} \qquad Y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0} \qquad Y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0} \qquad Y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1=0}$$

$$I1 = Y_{11}V_1 + Y_{12}V_2 \quad I2 = Y_{21}V_1 + Y_{22}V_2 \qquad \text{or} \qquad \begin{vmatrix} I1 \\ I2 \end{vmatrix} = \begin{vmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{vmatrix} * \begin{vmatrix} V1 \\ V2 \end{vmatrix}$$

Y-parameters describe how currents depend on voltages. These parameters are especially useful when two-port networks are connected in parallel, because admittances can be added directly.

For cascaded two-port networks, transmission or ABCD parameters are more useful. The ABCD relationship is:



$$A = \left.\frac{V_1}{V_2}\right|_{I_2=0} \qquad B = \left.\frac{V_1}{-I_2}\right|_{V_2=0} \qquad C = \left.\frac{I_1}{V_2}\right|_{I_2=0} \qquad D = \left.\frac{I_1}{-I_2}\right|_{V_2=0}$$

$$V1 = A * V_2 + B * -I_2 \quad I1 = C * V_2 + D * -I_2 \qquad \text{or} \qquad \begin{vmatrix} V1 \\ I1 \end{vmatrix} = \begin{vmatrix} A & B \\ C & D \end{vmatrix} * \begin{vmatrix} V2 \\ -I2 \end{vmatrix}$$

The objective of this experiment was to measure and calculate Z, Y, and ABCD parameters for different two-port networks and compare theoretical values with experimental measurements.

2. Execution

2.1Experiment Setup

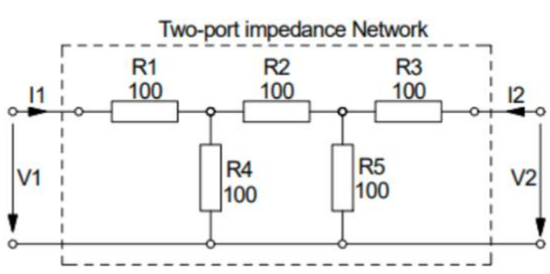
Workbench No.6

Used tools and instruments:

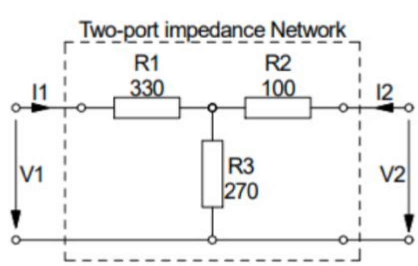
- Breadboard and toolbox
- Multimeter Tenma
- Elabo
- Oscilloscope
- Signal generator
- Passive probe
- BNC-Banana connector
- Various resistors
- 10 mH inductor
- 1 µF capacitors

2.2 Part 1: Two-port Z / Y Network – Setup

The objective of the first part was to determine the Z and Y-parameters of two different two-port networks. Two circuits were built on the breadboard. Circuit 1 consisted of five 100 Ω resistors. Circuit 2 consisted of 330 Ω, 100 Ω, and 270 Ω resistors. Both circuits were connected to a DC supply of approximately 5 V with a 1 kΩ load resistor at the output.



Circuit 1



Circuit 2

2.3 Part 1 : Two-port Z / Y Network - Execution and Results

To determine the Z-Parameters of circuit 1, Figure 1 was used as a reference. Z_{11} and Z_{22} were obtained by directly measuring the resistance at each respective port. For Z_{12} and Z_{21}, the opposite port was left open (setting I1 or I2 = 0), and the voltage across the terminal was measured using the Tenma multimeter. The corresponding current was then determined by measuring the voltage drop across a resistor and applying Ohm's law with the known resistance value. Once the currents from each circuit were summed, the Z-parameters were computed using the formulas from Figure 1. The results are presented below:

Z11	Z22	Z12	Z21
166.745	166.703	33.3509	33.3586

Additional measured values during this part were:

z11/z22			
I1	I2	V1	V2
12.687 mA	12.68 mA	2.1155 V	2.1138V

For Circuit 2, the admittance parameters were measured. The measured Y-parameters were:

circuit II impedance			
Y11	Y22	Y12	Y21
0.002491	0.00403	-0.00181	-0.00181

Lastly, we connect a 1kΩ resistor to the load and a 5V source at V1 then proceed to measure the voltages directly and the current. The voltage and current values were measured for both circuits.

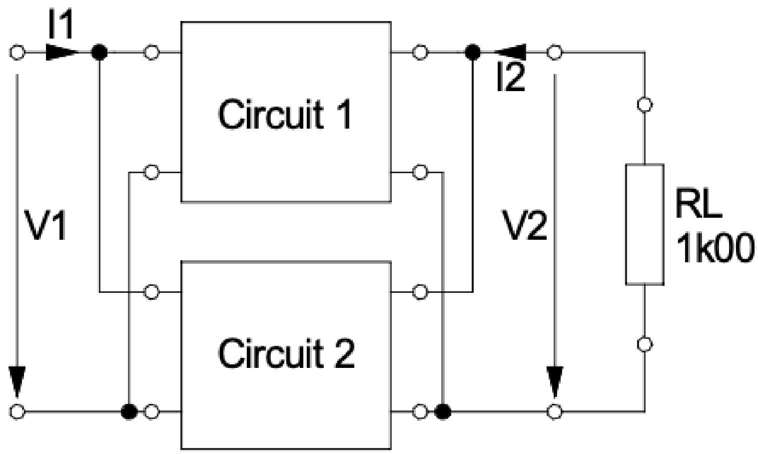
Measured Voltage and Current values of Circuit 1 and Circuit 2 are:

Circuit 1			
V1	V2	I1	I2
5.062	0.874	0.03058	-870uA
circuit 2			
V1	V2	I1	I2
5.069	2.2786	0.008456	-0.001828

2.4 Part 2 : Interconnection of Two-port Networks - Setup

The objective of this part was to investigate what happens when we connect two-port networks in parallel. Circuit 1 and Circuit 2 from Part 1 were connected in parallel by joining the corresponding input and output ports. A 1 kΩ load resistor was connected at the output.

In a parallel connection, the voltages across the two two-port networks are the same, while the currents split between the two networks. Because of this, admittance parameters are useful for combining the networks.



2.4 Part 2 : Interconnection of Two-port Networks - Setup

The objective of this part was to investigate the parallel connection of two two-port networks. Circuit 1 and Circuit 2 from Part 1 were connected in parallel and 1 kΩ load resistor was connected at the output port.

In a parallel connection, the voltages across the two two-port networks are the same, while the currents split between the two networks. So it is easier to use, admittance parameters.

2.5 Part 2 : Interconnection of Two-port Networks - Execution and Results

We measure the Z-parametres of the new Two-port. We do this exactly the same way we calculated in part 1.

Part2 interconnection of two-port networks			
Z11	Z22	Z12	Z21
127.56	108.53	38.0413	38.07703

A 5V DC supply was connected to V1, and a 1 kΩ load resistor R_L was placed across V2. The objective was to measure V1, V2, I1, and I2. V1 and V2 were measured straightforwardly using the Tenma directly across each port. I2 was found by measuring the voltage drop across R_L in the direction of I2 as shown in Figure 5, then applying Ohm's law with the known value of R_L. Finding I1 required a similar approach to the one used for the Z-parameters: the voltage drop across the first resistor in each circuit — taken in the direction of current flow — was measured and combined with Ohm's law to obtain the partial currents I1a and I1b, which were then added together to give the total I1. The results are presented below.

V1 (V)	V2 (V)	I1(A)	I2 (A)
5.031	1.366	0.03977	-0.00138

Voltage and current measurements with 1kΩ resistor load for parallel connection of circuits

2.6 Part 3: Complex Two-port Networks / Cascading – Setup

This part's objective is to observe how TPNs behave in cascade and in AC. The circuits contained capacitors, an inductor, and resistors. Since the circuit was supplied with an AC signal, all measured voltages and currents had to be treated as phasors.

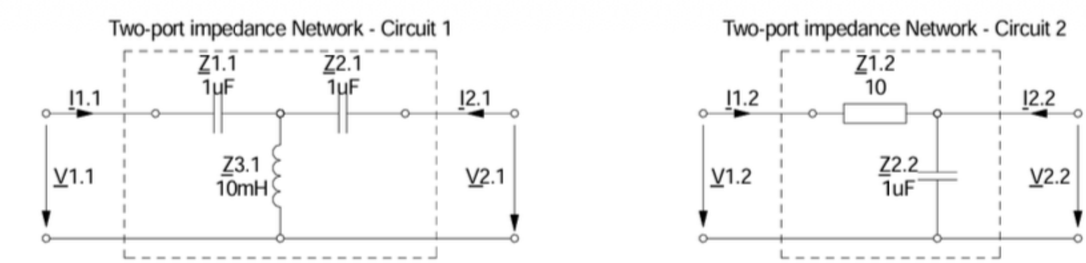
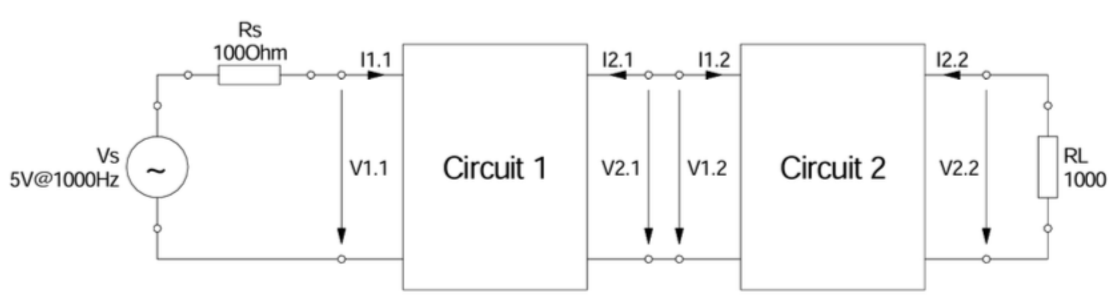


Figure 6: Impedance TPN circuits for experiment part 3



The signal generator was set to a sinusoidal signal at 1 kHz. The input voltage, output voltage, and phase shift were measured with the oscilloscope. The current was calculated using the voltage across the known resistors.

The input signal: Vpp = 5 V @1000Hz.

2.7 Part 3 : Complex Two-port Networks/Cascading - Execution and Results

Using the Tenma oscilloscope, the following quantities were measured: $\hat{v}_S$, $\hat{v}_{1.1}$, and $\hat{\imath}_{1.1}$ in circuit 1, and $\hat{v}_{2.2}$ and $\hat{\imath}_{2.2}$ in circuit 2. Channel 1 of the oscilloscope was connected across $\hat{v}_{1.1}$, with its ground tied to the source ground just below it. Channel 2 shared the same ground point, and its probe was repositioned across different nodes as indicated in Figure 7 to capture the required voltages. The currents were then derived from the previously measured voltages using Ohm's law: $\hat{\imath}_{1.1} = -\hat{v}_{1.1} / 100$ A, and $\hat{\imath}_{2.2} = -\hat{v}_{2.2} / 1000$ A.

	Magnitude	Phase relative to V_1.1
V_s	3.92 V	-46
V_1.1	2.48 V	0
I_1.1	3.8952 mA	0
V_2.2	1.18 V	-167
I_2.2	-0.00118 A	13

Here, V_1.1 was used as the reference phasor with phase angle 0^\circ. Since the output current direction is defined opposite to the measured load current direction, the sign of I_2.2 must be considered carefully.

The measured inductor value was L_S=10.389mH with R_S=4.4168, giving Z_3.1=65.439<86.13. The capacitors were measured around 0.9487mu F, 1.0799mu F, and 1.0820\mu F, giving capacitive impedances with phases close to -90, as expected. Since the resistor Z_1.2, source resistor,

	Z_11	Z_21	Z_31	Z_12	Z_22	Rs	RL
Polar	168.37<-89.77	149.52<-89.77	65.439< 86.13	10<0	147.28<-89.77	100<0	1000<0
Geometric	0.676-j168.37	0.600-j149.52	4.417+j65.29	10	0.591-j147.28	100	1000

3. Evaluation

3.1 Part 1 : Two-port Z / Y Network - Evaluation

First, the theoretical Z-parameters of Circuit 1 are calculated using the given resistor values. Circuit 1 is symmetrical, therefore: $Z_{11} = Z_{22}$

For circuit 1: $Z_{11} = Z_{22} = 100 + \frac{100\times200}{100+200}$ $Z_{11} = Z_{22} = 100 + \frac{20000}{300}$ $Z_{11} = Z_{22} = 100 + 66.67 = 166.67\ \Omega$

$Z_{12} = Z_{21} = 100 \times \frac{100}{100+200}$ $Z_{12} = Z_{21} = 33.33\ \Omega$

So the calculated impedance matrix for Circuit 1 is: $Z_1 = \begin{bmatrix} 166.67 & 33.33 \\ 33.33 & 166.67 \end{bmatrix}\ \Omega$

The measured values matrix: $Z_{1measured} = \begin{bmatrix} 166.745 & 33.3509 \\ 33.3586 & 166.703 \end{bmatrix}\ \Omega$

The measured and calculated values are almost identical. The small difference is due to resistor tolerances, resistance in the breadboard, and multimeter reading uncertainty.

For circuit 2, the theoretical impedance are:

$Z_{11} = 330 + 270 = 600\Omega.$

$Z_{22} = 100 + 270 = 370\ \Omega$

$Z_{12} = Z_{21} = 270\ \Omega$

$Z_2 = \begin{bmatrix} 600 & 270 \\ 270 & 370 \end{bmatrix}\ \Omega$

The Y-parameters are obtained by simply inverting the Z-matrix:

det(Z_2)=600 .370 – 270.270

det(Z_2)=222000-72900=149100

$Y_2 = \frac{1}{149100}\begin{bmatrix} 370 & -270 \\ -270 & 600 \end{bmatrix} = \begin{bmatrix} 0.002481 & -0.001811 \\ -0.001811 & 0.004024 \end{bmatrix}$S

The calculated and measured values are very close. This confirms that the measurement method was correct.

Now the measured values of V_1, V_2, I_1, and I_2 are verified using the measured Z-parameters of Circuit 1.

$V_1 = Z_{11}I_1 + Z_{12}I_2 \qquad V_2 = Z_{21}I_1 + Z_{22}I_2$

$I_1 = 0.03058A$

$I_2 = -0.000870A$

$Z_{11} = 166.745\Omega$

$Z_{12} = 33.3509\Omega$

$Z_{21} = 33.3586\Omega$

$Z_{22} = 166.703\Omega$

Then we plug in the equation: $V_1 = 166.745(0.03058) + 33.3509(-0.000870)$

$V_1 = 5.1007 - 0.0290 = 5.0717V$

$V_2 = 33.3586(0.03058) + 166.703(-0.000870)$

$V_2 = 1.0200 - 0.1450 = 0.8750V$

The measured values are: $V_{1measured} = 5.062V \quad V_{2measured} = 0.874V$

Comparison of circuit 1 Calculated and Measured Z parameters:

	calculated	Measured
Z_11	166.67 Ω	166.745 Ω
Z_22	166.67 Ω	166.703 Ω
Z_12	33.33 Ω	33.3509 Ω
Z_21	33.33 Ω	33.3509 Ω

The measured impedance parameters are extremely close to the calculated values. . The small deviations most likely caused by resistor tolerance, breadboard resistance, and multimeter uncertainty.

	calculated	Measured
Y_11	0.002481 S	0.002491 S

Y_22	0.004024 S	0.004030 S
Y_12	-0.001811 S	-0.001810 S

Comparison of circuit 2 Calculated and Measured Y parameters:

The measured admittance parameters of Circuit 2 are also very close to the theoretical values. This confirms that the Y-parameter measurement was accurate.

3.2 Part 2: Interconnection of Two-port Networks – Evaluation

For a parallel connection of two two-port networks, the admittance matrices can be added directly: $Y_{eq} = Y_1 + Y_2$

Circuit 1 was measured in Z-parameters, so first Z_1 must be inverted to obtain Y_1.

$$Z\_1=\begin{bmatrix}166.745 & 33.3509\\ 33.3586 & 166.703\end{bmatrix}\,\Omega.\;\; Y_1 = \frac{1}{\det(Z_1)}\begin{bmatrix}Z_{22} & -Z_{12}\\ -Z_{21} & Z_{11}\end{bmatrix}=Y_1=\begin{bmatrix}0.00625 & -0.00125\\ -0.00125 & 0.00625\end{bmatrix}\mathrm{S}$$

Circuit 2 measured admittance is: $Y_2=\begin{bmatrix}0.002491 & -0.00181\\ -0.00181 & 0.00403\end{bmatrix}\mathrm{S}$

Equivalently: $Y_{eq} = Y_1 + Y_2$

$$Y_{eq} = \begin{bmatrix}0.00625+0.002491 & 0.00125-0.00181\\ -0.00125-0.00181 & 0.00625+0.00403\end{bmatrix}=\begin{bmatrix}0.008741 & -0.00306\\ -0.00306 & 0.01028\end{bmatrix}$$

Z parameter will be simply just inverted matrix of Y

$$\det(Y_{eq}) = 0.00008051$$

$$Z_{eq} = \frac{1}{0.00008051}\begin{bmatrix}0.008741 & -0.00306\\ -0.00306 & 0.01028\end{bmatrix}=\begin{bmatrix}127.68 & 38\\ 38 & 108.57\end{bmatrix}$$

$$Z_{eqmeasured} = \begin{bmatrix}127.68 & 38\\ 38 & 108.57\end{bmatrix}$$

The calculated and measured values match very well.

Now the measured voltages are verified using the measured Z-parameters of the parallel circuit:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 127.56 & 38.0413 \\ 38.07703 & 108.53 \end{bmatrix} \begin{bmatrix} 0.03977 \\ -0.00138 \end{bmatrix}$$

$V_1 = (127.56)(0.03977) + (38.0413)(-0.00138) = 5.0726 - 0.0525 = 5.0201V$

$V_2 = (38.07703)(0.03977) +)(108.53)(-0.00138)$=1.5147-0.1498=1.3649V

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 5.0201 \\ 1.3649 \end{bmatrix} V$$

Measured Values: $V_{1measured} = 5.031V.\ \ V_{2measured} = 1.366V$

The calculated values are almost equal to the measured values. Therefore, the equivalent Z-parameters of the parallel connected two-port network are verified.

It's not always possible to find the Z-parameters in a series connection of TPNs by direct addition of the impedances. The two conditions for series connections are that the input current of the first circuit is equal to the output of the second, and that the voltages sum to the sources. The first condition of currents can already be seen as violated, since if there is an input current I1, it would be split between resistors R4 and R5, and the load of circuit 2, so the output current of circuit 1 would already not match it's input. Due to the configuration of these resistors, it is more complicated to find the equivalent series impedances of the two circuits.

3.3 Part 3: Complex Two-port Networks / Cascading – Evaluation

In Part 3, the two-port were build using capacitors and an inductor, therefore complex impedance values had to be used. The purpose of this part was to determine the Z-parameters of each circuit, convert them to ABCD parameters, cascade the networks, and verify the measured input voltage and current using the measured output voltage and current.

For circuit 1

$Z_{11} = Z_{1.1} + Z_{3.1}$ Z_{11}=(0.676-j168.37) + (4.417 + j65.29)= 5.093-j103.08

$Z_{22} = Z_{2.1} + Z_{3.1}$ $Z_{22} = (0.600 - j149.52) + (4.417 + j65.29)$ =5.017-j84.23

$Z_{12} = Z_{21} = Z_{3.1}$ $Z_{21} = Z_{21} = Z_{3.1}$=4.417 + j65.29

For circuit 2

$Z_{11} = Z_{1.2} + Z_{2.2}$ $Z_{11} = 10 + (0.591 - j147.28) = 10.591 - j147.28$

$$Z_{12} = Z_{21} = Z_{22} = Z_{2.2} = 0.591 - j147.28$$

The general conversion from Z-parameters to ABCD parameters is:

$$A = \frac{Z_{11}}{Z_{21}}, B = \frac{Z_{11}Z_{22} - Z_{21}Z_{21}}{Z_{21}}, C = \frac{1}{Z_{21}}, D = \frac{Z_{22}}{Z_{21}}$$

For two cascaded networks, the total ABCD matrix is:

$$ABCD_{total} = ANCD_1 ANCD_2$$

We have to convert Z parameters in ABCD parametres which is equal to: $\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{Z_{11}Z_{22} - Z_{12}Z\_21}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix}$

Thus, the final impedance matrices for both circuits are:

$$Z_1 = \begin{bmatrix} 5.093 - j103.08 & 4.417 + j65.29 \\ 4.417 + j65.29 & 5.017 - j84.23 \end{bmatrix}$$ For circuit 1

$$Z_2 = \begin{bmatrix} 10.591 - \mathrm{j}147.28 & 0.591 - \mathrm{j}147.28 \\ 0.591 - \mathrm{j}147.28 & 0.591 - \mathrm{j}147.28 \end{bmatrix}$$ For circuit 2

For circuit 1 ABCD Parametres:

$$A_1 = \frac{5.093 - j103.08}{4.417 + j65.29}$$

$$A_1 = -1.566 - j0.184$$

$$B_1 = \frac{(5.093 - j103.08)(5.017 - j84.23) - (4.417 + j65.29)(4.417 + j65.29)}{4.417 + j65.29}$$

$$B_1 = -27.77 + j65.72$$

$$C_1 = \frac{1}{4.417 + j65.29}$$

$$C_1 = 0.00103 - j0.01525$$

$$D_1 = \frac{5.017 - j84.23}{4.417 + j65.29}$$

$$D_1 = -1.279 - j0.163$$

For circuit 2 ABCD Parametres

$$A_2 = \frac{10.591 - j147.28}{0.591 - j147.28}$$

$$A_2 = 1.000 + j0.0679$$

$$B_2 = \frac{(10.591 - j147.28)(0.591 - j147.28) - (0.591 - j147.28)(0.591 - j147.28)}{0.591 - j147.28}$$

$$B_2 = 10.00$$

$$C_2 = \frac{1}{0.591 - j147.28}$$

$$C_2 = 0.0000272 + j0.00679$$

$$D_2 = \frac{0.591 - j147.28}{0.591 - j147.28}$$

$$D_2 = 1$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix}$$

$$\mathrm{ABCD}_1 = \begin{bmatrix} -1.566 - \mathrm{j}0.184 & -27.77 + \mathrm{j}65.72 \\ 0.00103 - \mathrm{j}0.01525 & -1.279 - \mathrm{j}0.163 \end{bmatrix}$$

$$ABCD_2 = \begin{bmatrix} 1.000 + j0.0679 & 10.00 \\ 0.0000272 + j0.00679 & 1 \end{bmatrix}$$

After calculating the multiplication of this two matrix the input voltage and current can be calculated from: $\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$

Total ABCD matrix is: $\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} -2.0005 - j0.4771 & -43.43 + j63.88 \\ 0.00314 - j0.02387 & -1.2687 - j0.3155 \end{bmatrix}$

$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$ Now we expand this $V_1 = AV_2 + BI_2$ $I_1 = CV_2 + DI_2$

measured output voltage was $V_2 = 1.18\angle - 167°$This in quadratic form is $V_2 = 1.150 - j0.265$

Since the load resistor is 1k ohm , the output current magnitude is $I_2 = \frac{1.18}{1000} = 0.00118 \, which \, can \, be \, written \; -I_2 = 0.00118\angle - 167° => \; -I_2 = -0.001150 - j0.000265$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} -2.0005 - j0.4771 & -43.43 + j63.88 \\ 0.00314 - j0.02387 & -1.2687 - j0.3155 \end{bmatrix} \begin{bmatrix} -1.150 - j0.265 \\ -0.001150 - j0.000265 \end{bmatrix}$$

V1 will be equal to:

$V_1 \approx 2.240 + j1.018\, V$=$V_1 \approx 2.46\angle 24.43°\, V$

I1 will be equal to:

$$I_1 = (0.00314 - j0.02387)(-1.150 - j0.265) + (-1.2687 - j0.3155)(-0.001150 - j0.000265)$$

$I_1 \approx -0.00857 + j0.02731\, A$ $I_1 \approx 0.0286\angle 107.42°\, A$

In this part, larger deviations are expected compared to the DC parts of the experiment. This is because AC phasor measurements include both magnitude and phase errors. Small phase measurement errors can cause large deviations after converting to rectangular complex form and multiplying matrices. In addition, the capacitors and inductor are not ideal components. They include internal resistance and parasitic effects, which influence both magnitude and phase.

4.Conclusion

In Part ,the experiment concentrated on measuring the Z-parameters of Circuit 1 and the Y-parameters of Circuit 2 and then comparing against theoretically calculated values. The results matched closely, confirming that impedance and admittance matrix models accurately describe two-port network behavior. This part also reinforced how Z and Y parameters can be used to express Ohm's law for two-port networks.

In Part 2, the two circuits were connected in parallel. The equivalent Z-parameters were obtained by converting Circuit 1 to Y-parameters, summing the admittance matrices, and converting back to Z-parameters. The calculated values matched with the directly measured ones, demonstrating that admittance parameters are particularly well-suited for analyzing parallel two-port networks.

Part 3 extended the analysis to cascaded two-port networks with reactive components and AC signals, using ABCD parameters. This part proved more challenging, as all quantities were phasors and component non-idealities introduced additional error. The most significant sources of inconsistency were phase measurement uncertainty, output current sign convention, and accumulated errors in complex-number calculations particularly due to the high tolerance of reactive components and equipment resolution. Nevertheless, the core method was valid: cascaded two-port networks can be effectively analyzed by multiplying their ABCD matrices.

Overall, the experiment demonstrated that two-port network parameters provide a powerful and flexible framework for circuit analysis. Z-parameters are best suited for impedance modeling, Y-parameters for parallel configurations, and ABCD parameters for cascaded networks.

5. Appendix

Circuit 1 Impedance				
	Z11	**Z22**	**Z12**	**Z21**
Value	166.745 Ω	166.703 Ω	33.3509 Ω	33.3586 Ω

Circuit 1 Extra Data				
	I1	**I2**	**V1**	**V2**
Value	12.687 mA	12.68 mA	2.1155 V	2.1138 V

Circuit 2 Admittance				
	Y11	**Y22**	**Y12**	**Y21**
Value	0.002491 S	0.00403 S	-0.00181 S	-0.00181 S

Circuit 1 with RL = 1kΩ				
	V1	**V2**	**I1**	**I2**
Value	5.062 V	0.874 V	0.03058 A	-870 μA

Circuit 2 with RL = 1kΩ				
	V1	**V2**	**I1**	**I2**
Value	5.069 V	2.2786 V	0.008456 A	-0.001828 A

6. References

Uwe, P. (2024, 11 21). Retrieved from Uwe Pagel - Constructor University Natural Sciences Laboratory